

Machine Learning for Biology and Health

CSCI 1851
Spring 2026

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Tuesday



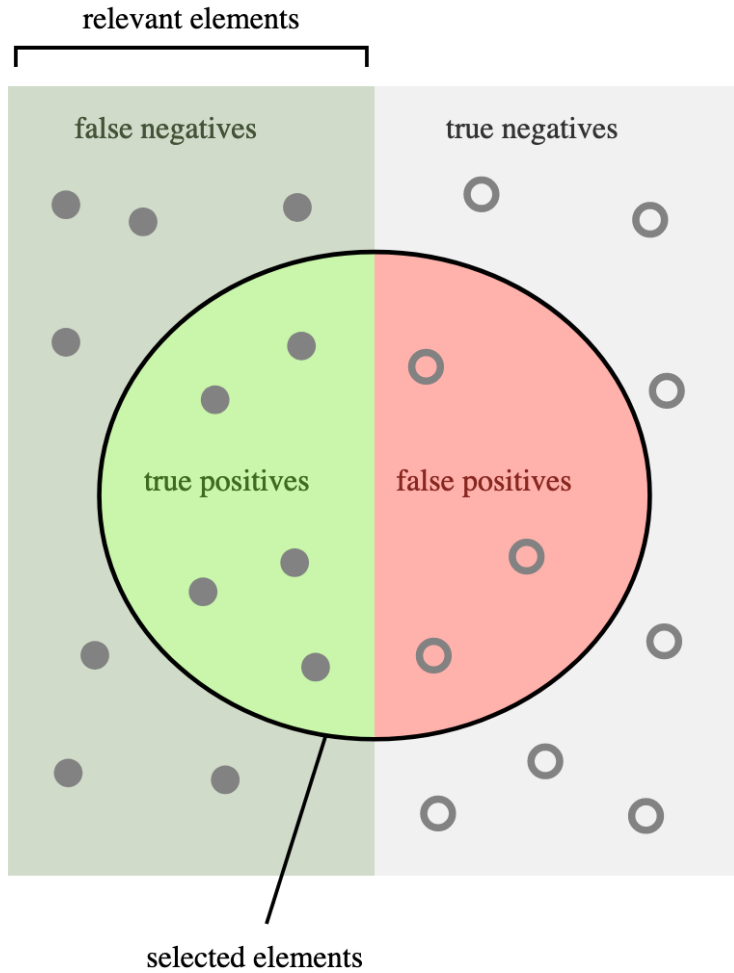
Cross Entropy Loss (for Binary classification)

$y = \text{true label of class (0 or 1)}$

$p = \text{predicted probability of class 1}$

$$\mathcal{L}(w, b) = \frac{1}{n} \sum_{i=1}^n [-y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)]$$

When accuracy is not enough.



Today's goal - Learn about decision trees and random forests for classification

(1) Decision Tree

(2) Loss Function

(3) Learning Algorithm

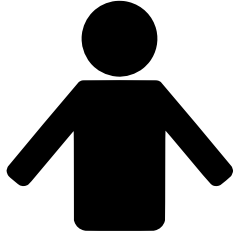
(4) Random Forest

(5) Class Activity: Trees and Forests

(6) Random Forests versus Decision Trees

Decision Tree

Our problem: Heart disease classification



Input: $\mathbb{X}$

age,cholesterol,resting_bp,max_hr
45,180,120,170
48,185,118,172
50,190,125,165
52,200,130,160
55,210,135,158
57,220,140,150
58,235,142,152
60,240,145,148
62,250,150,140
65,260,155,138



Function: f



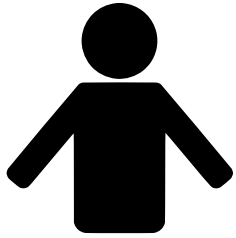
$$f(\mathbb{X}) \rightarrow \mathbb{Y}$$

Target: $\mathbb{Y}$



heart_disease
0
0
0
0
0
0
1
1
1
1
1
1

Simpler example: Learning function f



Input: $\mathbb{X}$

“Age, Cholestrol”

Binary classification

Linear decision boundary

Target: $\mathbb{Y}$



“Has heart disease?”

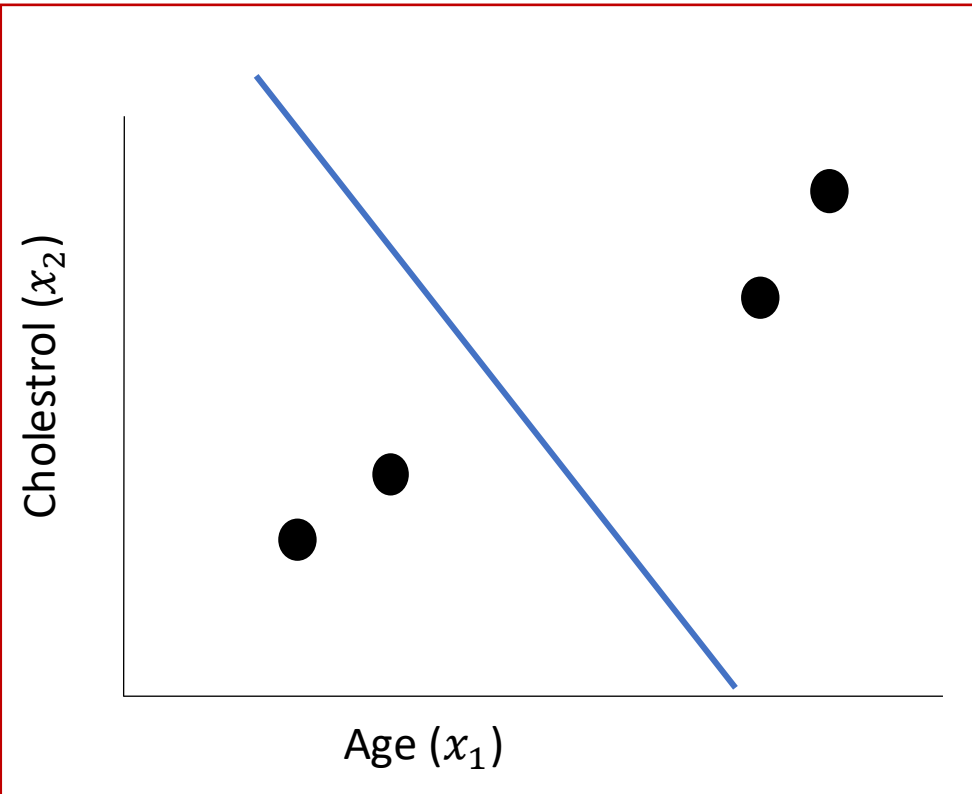
$\mathbb{X} \in \mathbb{Z}$

$x^{(1)}$ 45, 180

$x^{(2)}$ 48, 185

$x^{(3)}$ 62, 250

$x^{(4)}$ 65, 260



$y^{(1)}$ 0

$y^{(2)}$ 0

$y^{(3)}$ 1

$y^{(4)}$ 1

$\mathbb{Y} \in \{0,1\}$

What if we designed a flow chart for this?

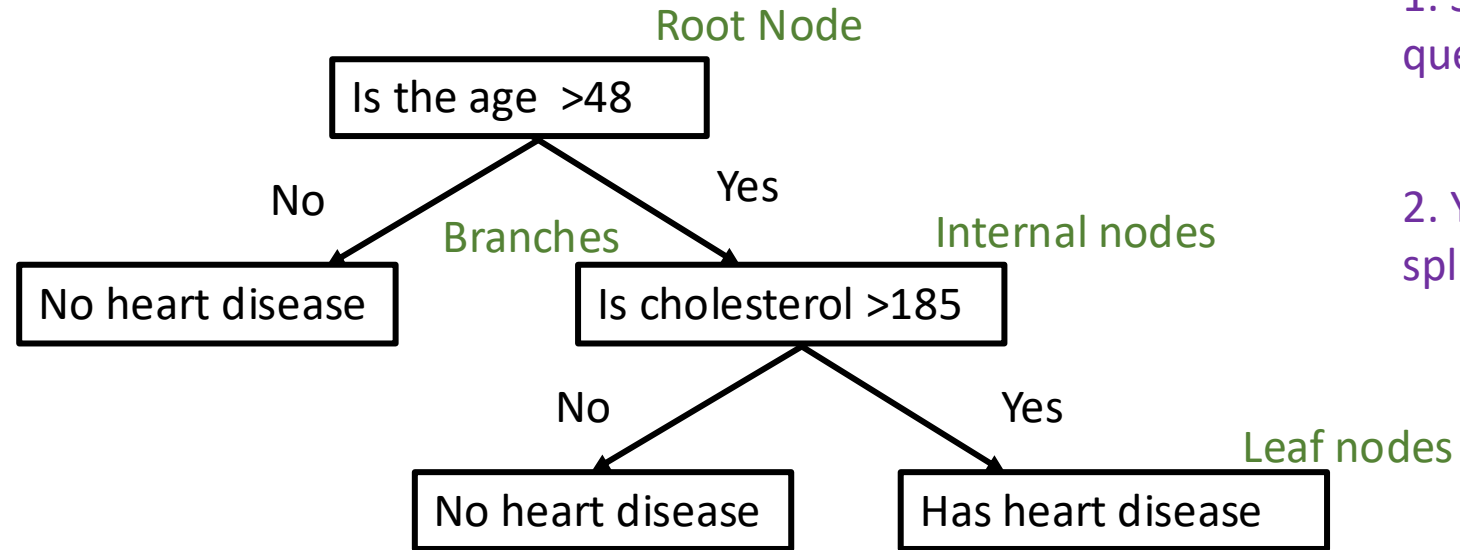
“Age, Cholestrol”

$x^{(1)}$ 45, 180

$x^{(2)}$ 48, 185

$x^{(3)}$ 62, 250

$x^{(4)}$ 65, 260



1. Starts with a question

2. Yes/No questions split the data

3. Reaches final decision

Decision Tree

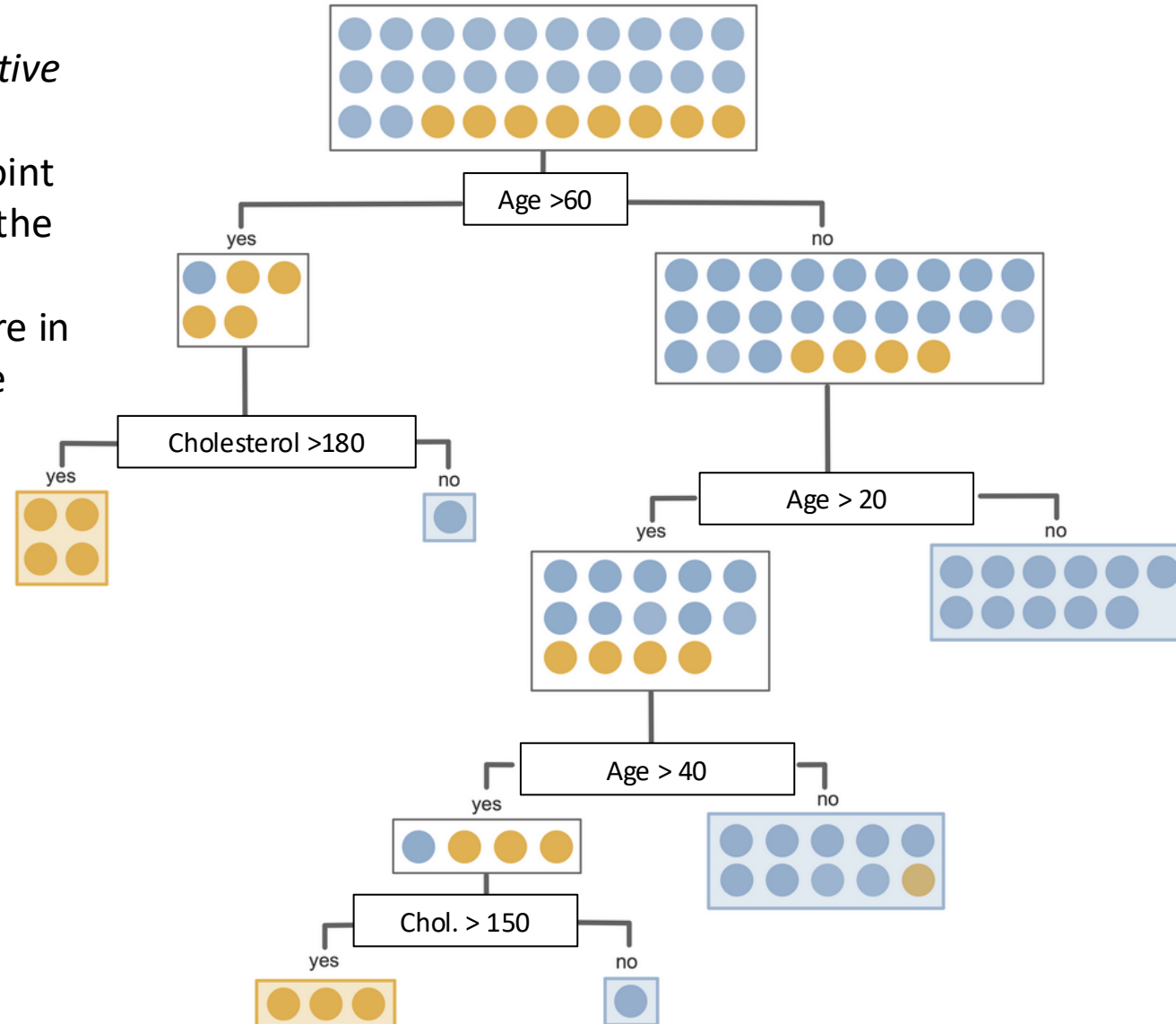
- A decision tree maps out decisions and their outcomes
- Starts with a root node and branches out into decisions
- Type of non-parametric model

Classification Tree

Can you do regression?

Decision tree with n samples and two classes

Binary Classification: A *positive class probability response* prediction for a new data point can be computed based on the proportion of positive class training observations that are in the corresponding leaf node



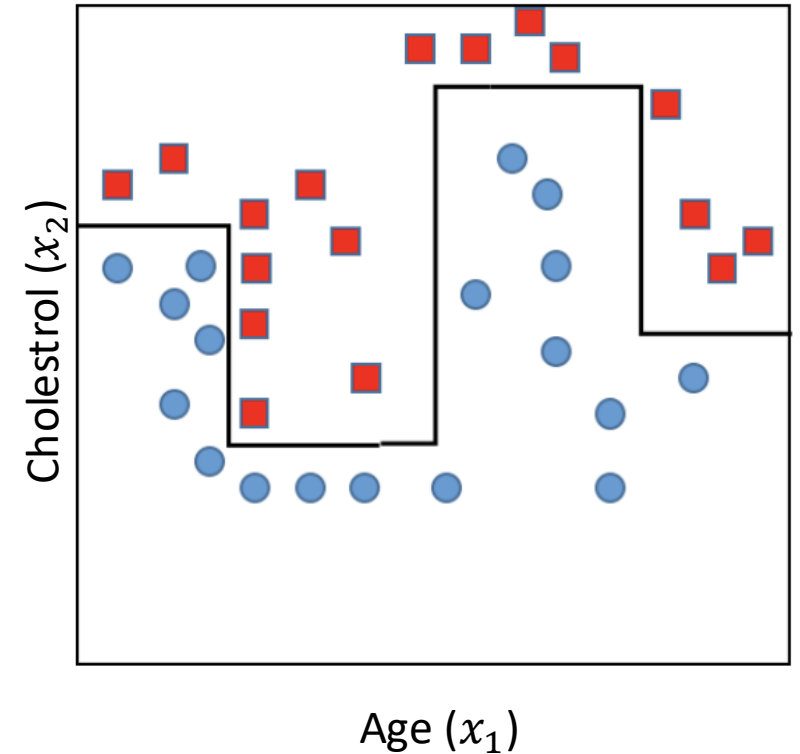
Any questions?



Decisions are not calculated – just for demonstration

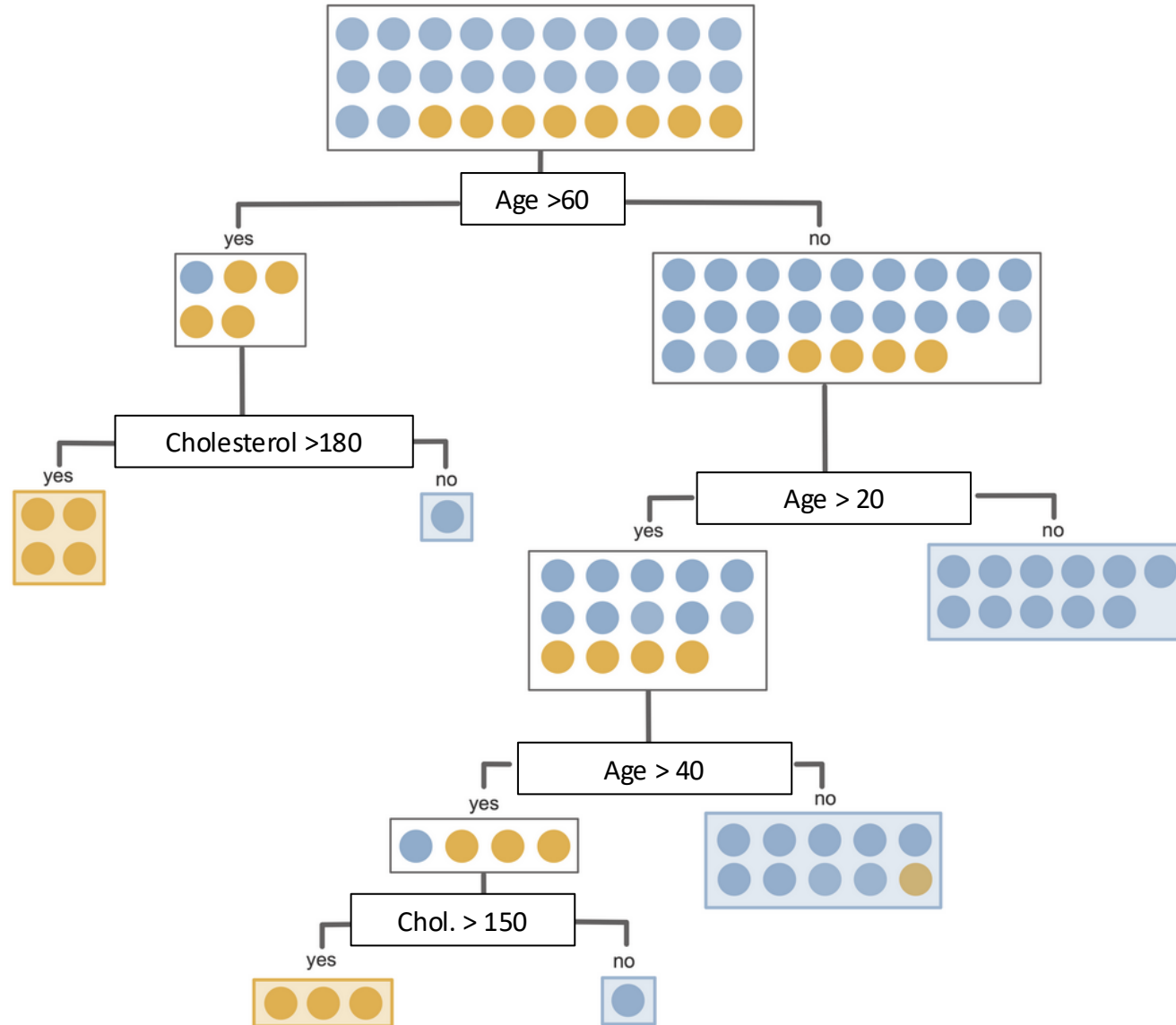
Why use decision trees?

- Can handle non-linear spaces using multiple decision boundaries
- Very interpretable



Loss Function (or Splitting Criteria)

Pure versus impure node



Gini Impurity

Goal: To find the split that leads to the most pure/least impure child nodes, as measured by the Gini impurity.

The Gini impurity of a set of observations is defined as the probability that two randomly chosen observations from the node are in different classes

Gini impurity = $P(\text{two random obs in different classes})$.

What will be this value
for a pure node?

What will be this value
for a maximally impure
node?

Gini Impurity

Gini impurity = $P(\text{two random obs in different classes})$.

Gini impurity = $1 - P(\text{two random obs in the same class})$.

Gini impurity = $1 - \left(P(\text{both pos class}) + P(\text{both neg class}) \right)$,

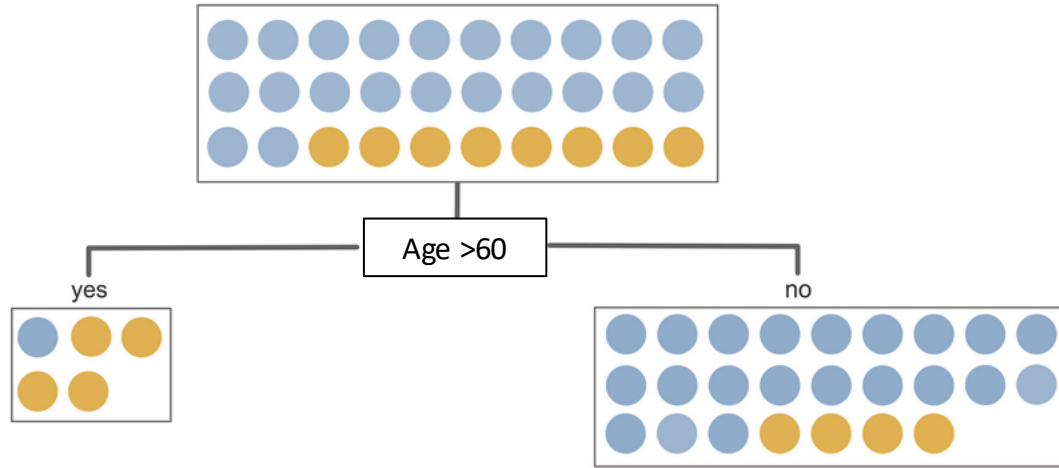
$p_1 = P(\text{a randomly selected observation is in the positive class})$.

$$\text{Gini impurity} = 1 - p_1^2 - p_0^2,$$

Any questions?



Gini Impurity for the split



When to stop splitting?

Maximum depth: The maximum depth of the tree (i.e., the maximum number of sequential splits allowed).

Minimum node size: The minimum number of observations required to split a parent node.

Any questions?



Learning Algorithm

Classification and Regression Tree (CART) Algorithm

Start with a top-level parent node that contains all the training data observations. Then:

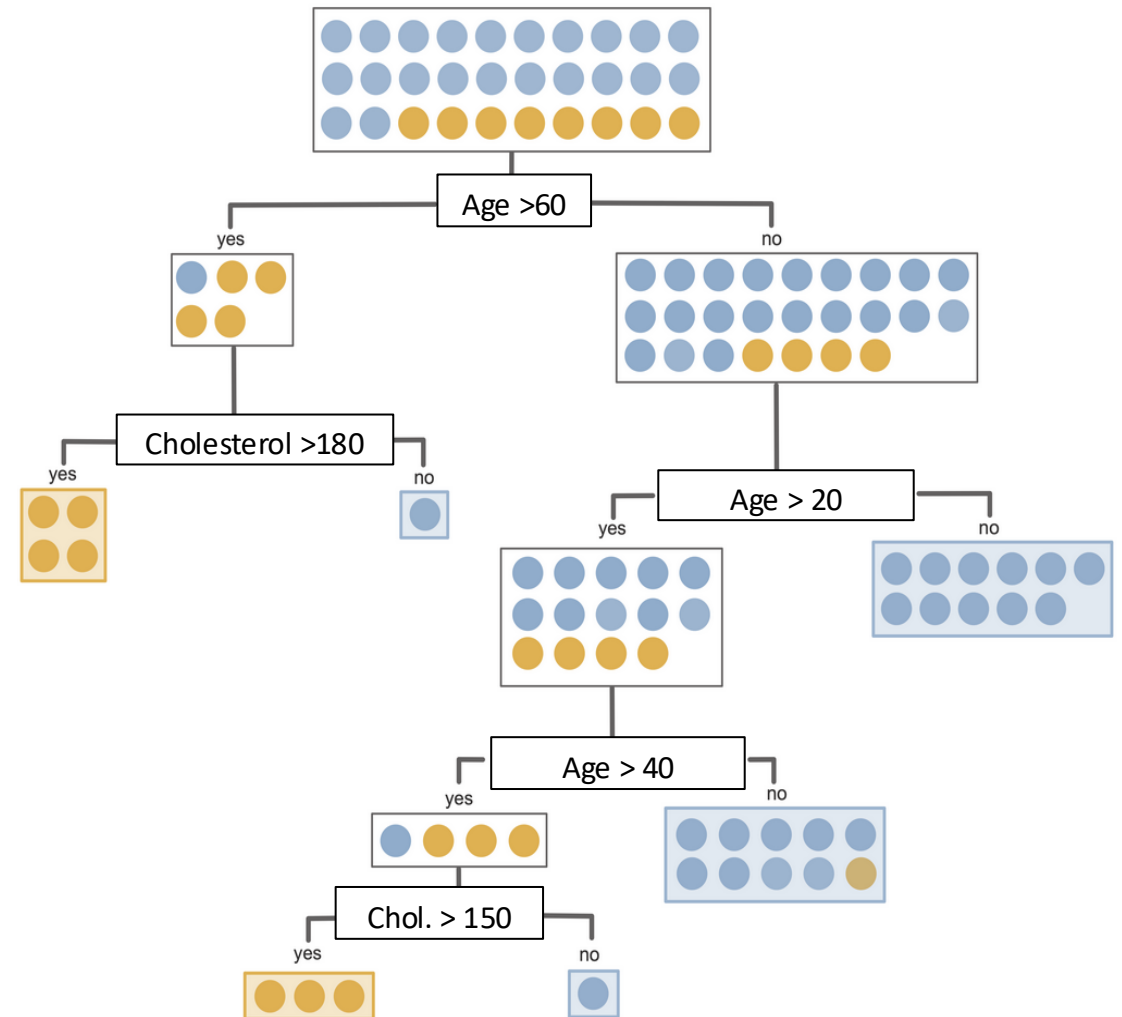
1. Conduct a **greedy search** for the “best” split of the parent node by
 1. Identifying a range of reasonable possible splits each consisting of a binary question involving a feature and threshold.
 2. For each potential split option identified in the previous step, compute the variance of the responses (for continuous responses) or Gini impurity (for binary responses) in each of the resulting child nodes.
 3. Identify which of the potential splits leads to the lowest variance measure (for continuous responses) or Gini impurity measure (for binary responses) of the child nodes. This is the “best” split.
 2. Implement the best split identified in the previous step to create two child nodes.
 3. For each resulting child node that does not meet a prespecified stopping criteria, repeat steps 1 and 2 (i.e., each child node is now the parent node and will be split into subsequent child nodes).
- If a child node meets the prespecified stopping criteria do not split the child node. A child node that is not split further is called a “**leaf node**” or “**terminal node**”.
4. Continue computing sequential splits until all child nodes meet the prespecified stopping criteria.

Making predictions on test set

To generate predictions for new observations,

Identify the relevant leaf node for the new observation

Compute the average response of the training observations in the leaf node (for continuous responses) or the proportion of positive class training observations in the leaf node (for binary responses).



Any questions?

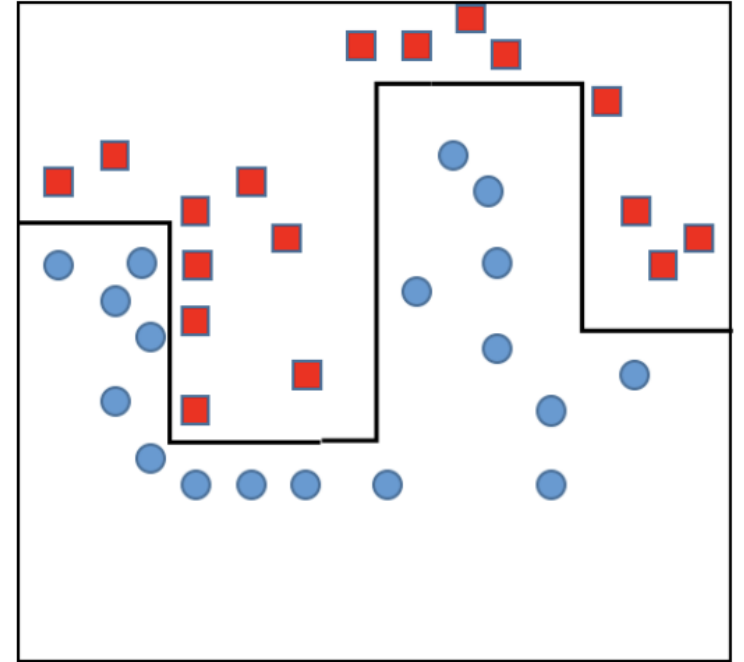


Why use decision trees?

- Can handle non-linear spaces using multiple decision boundaries
- Very interpretable
- Automatically handles decision-making

But...

- Prone to overfitting
- Can be affected by noise



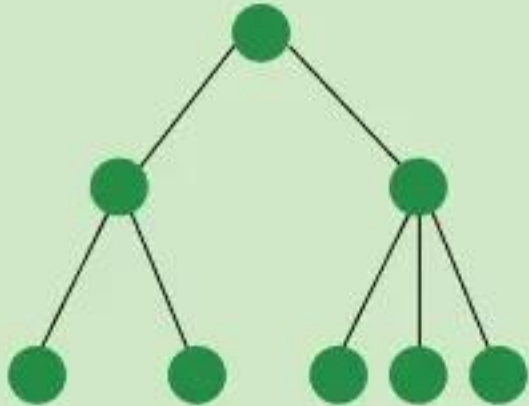
Random Forest

Random Forest Model

Ensemble learning

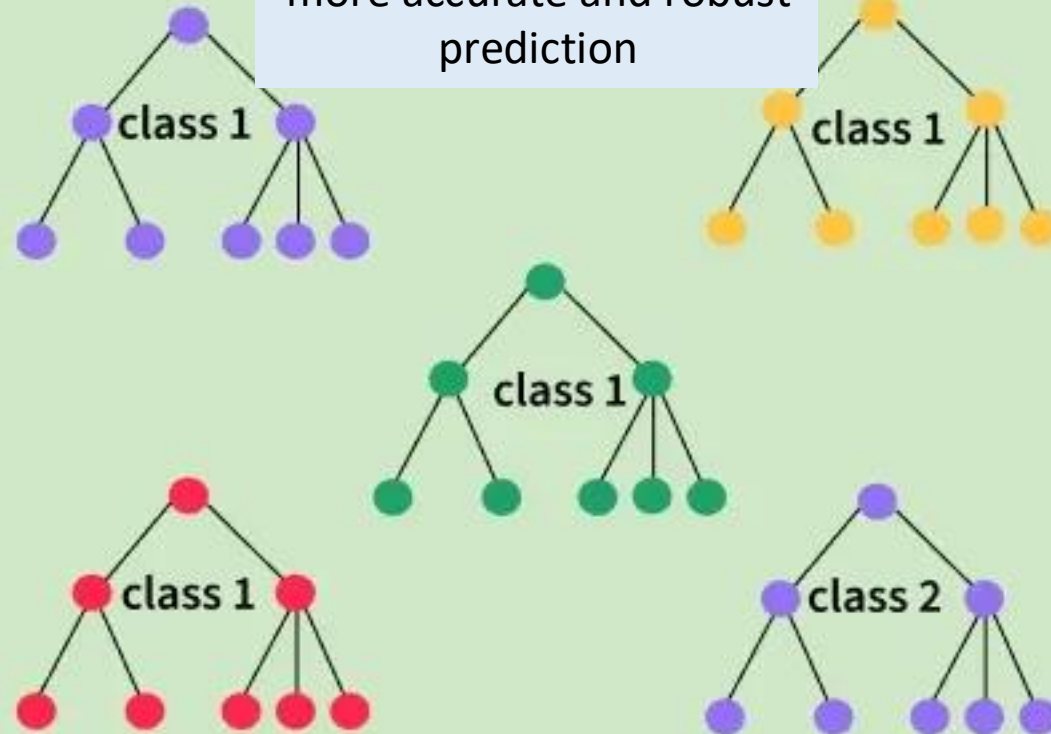
Single Decision Tree

Prediction from a single decision path



Random Forest

Ensemble of trees for more accurate and robust prediction



Random Forest Learning Algorithm

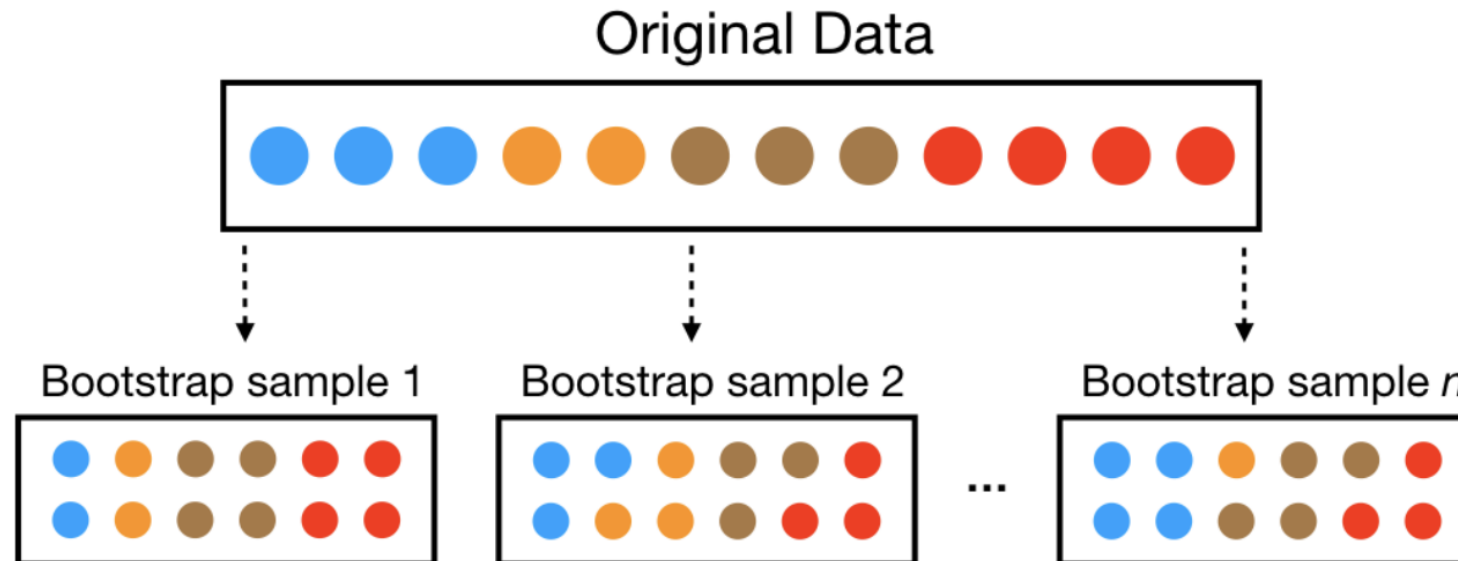


Randomness is introduced into each tree by:

- Training each tree using a different **random bootstrap sample** of the training data.

Bootstrapping

Bootstrapping is a statistical sampling technique that creates subsets of data through **random sampling with replacement**.



Random Forest Learning Algorithm

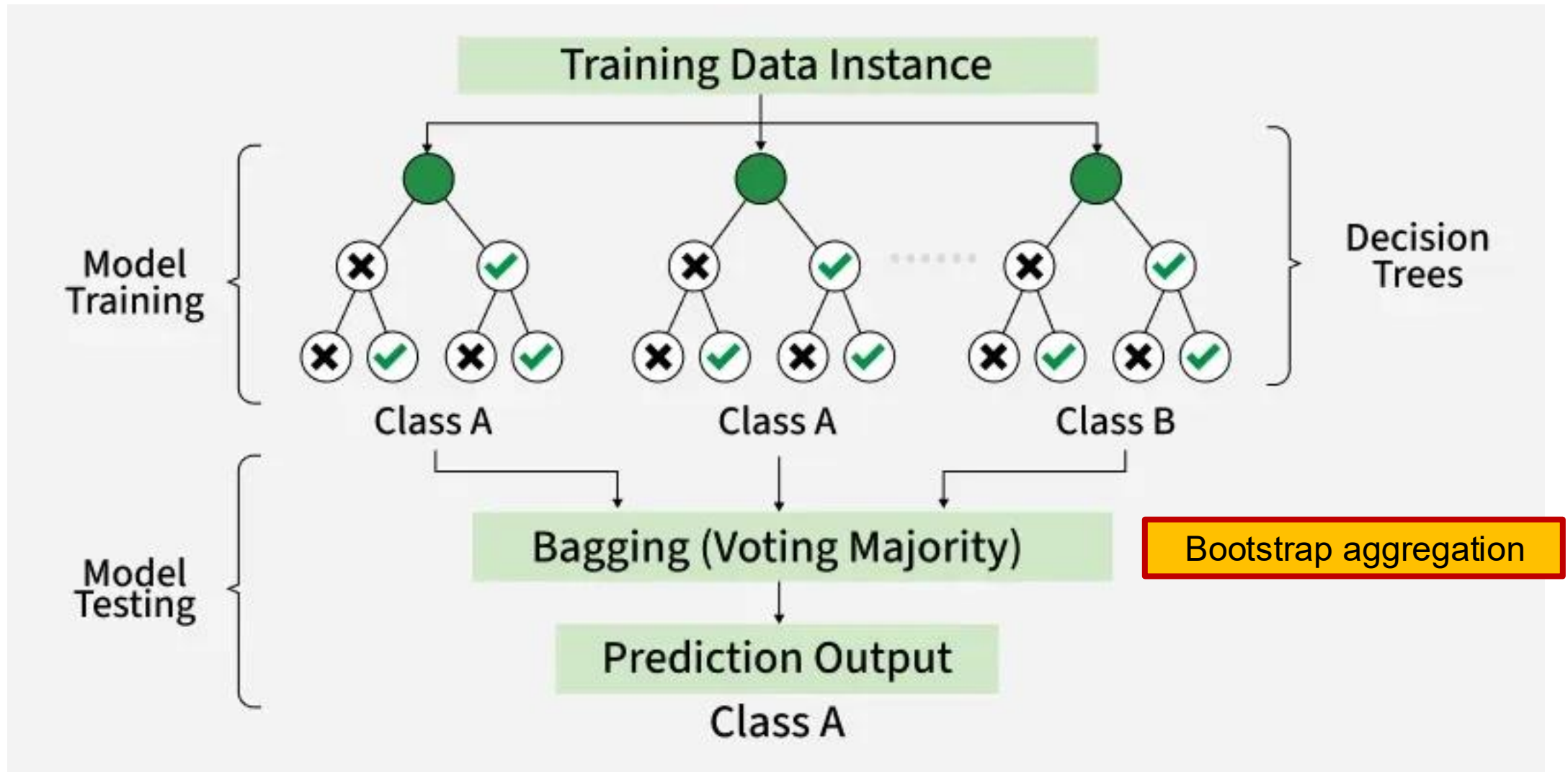


Randomness is introduced into each tree by

- Training each tree using a different random bootstrap sample of the training data.
- Considering a different random subset of the predictor variables for each node split.

This option can be implemented for the decision tree as well

Random Forest Learning Algorithm



Class activity

Decision Trees and Random Forests

Property	Random Forest	Decision Tree
Nature	Ensemble of multiple decision trees	Single Decision Tree
Interpretability		
Overfitting		
Training Time		
Stability to change		
Predictive Time		
Performance on large datasets		
Handling Outliers/Noise		
Feature Importance		

Property	Random Forest	Decision Tree
Nature	Ensemble of multiple decision trees	Single Decision Tree
Interpretability	Less interpretable due to ensemble nature.	Highly interpretable.
Overfitting	Due to ensemble averaging it is less prone to overfitting.	More prone to overfitting specially in case of deep trees.
Training Time	Since multiple trees are constructed, training time becomes more, and training speed becomes less.	A single tree needs to be built and trained, hence faster in comparison.
Stability to change	Since overall average is taken due to ensemble, it is more stable to change.	It becomes quite sensitive to variation in data.
Predictive Time	Multiple predictions, hence longer prediction time and slower prediction speed.	Faster prediction as compared to random forest, since a single prediction is made.
Performance on large datasets	Generally performs well on large datasets.	It can perform well on small and large dataset as well.
Handling Outliers	Due to ensemble averaging more robust to outliers/noise	It is more susceptible to outliers/noise
Feature Importance	Do not provide feature score directly rather uses ensemble to decide feature score.	Provide feature score directly which are less reliable.

Recap - Learn about decision trees and random forests for classification

(1) Decision Tree

(2) Loss Function

(3) Learning Algorithm

(4) Random Forest

(5) Class Activity: Trees and Forests

(6) Random Forests versus Decision Trees

Homework Reading:

[Chapter 12 in Veridical Data Science Book](#)

Sections 12.1, 12.2, 12.3, 12.4

Also please go over:

Very nice [visual explanation](#) using
Entropy for splitting

Wrap up



What was the clearest point today?

What was the muddiest point today?

